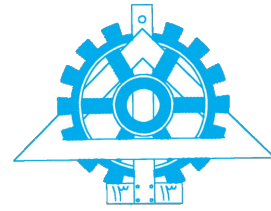




University of Tehran
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Machine Learning

Final Project

Names

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Abstract

This project presents a complete machine learning pipeline for multilingual speech language identification using classical machine learning methods. The dataset consists of 720 one-minute audiobook recordings in four languages: German, Italian, Korean, and Spanish. Several preprocessing steps were applied, including resampling, silence removal, and feature standardization, to ensure data consistency and quality.

A total of 46 acoustic features were extracted from each audio sample, including Mel-Frequency Cepstral Coefficients (MFCCs), spectral centroid, zero-crossing rate, and RMS energy. These features provide a comprehensive representation of spectral, temporal, and energy-related speech characteristics.

In the supervised learning stage, five classification models were implemented and evaluated: Logistic Regression, Support Vector Machine (RBF kernel), Multi-Layer Perceptron, K-Nearest Neighbors, and Random Forest. The results showed extremely high performance, with four models achieving 100% accuracy on the test set and Random Forest achieving 99%. These results demonstrate that the extracted features provide strong discriminative power for language identification.

In addition to classification, unsupervised clustering was performed using K-Means, Hierarchical Clustering, Gaussian Mixture Models, and DBSCAN. Among the tested methods, Hierarchical Clustering achieved the best overall performance in terms of cluster purity and external validation metrics. However, clustering results indicated that although meaningful structure exists in the feature space, languages are not perfectly separable in an unsupervised setting.

Overall, this project demonstrates that well-designed acoustic features combined with traditional machine learning algorithms can achieve highly effective multilingual speech classification, while also providing insight into the intrinsic structure of language representations in feature space.

Data Cleaning and Preprocessing

After collecting the multilingual audiobook recordings, several preprocessing steps were applied to ensure consistency and improve feature quality.

All audio files were standardized to a sampling rate of **22,050 Hz** to maintain uniform temporal and spectral resolution across recordings. Using a fixed sampling rate ensures that extracted frequency-based features are comparable between different samples and languages.

Silence trimming was performed to remove non-informative segments such as pauses at the beginning and end of recordings. A threshold-based method was used to eliminate low-energy regions. This step improves feature reliability by focusing only on active speech content and preventing silence from distorting statistical feature summaries.

The final dataset consisted of **720 audio samples**, each representing one speech recording labeled by language.

Feature Extraction

To transform raw audio signals into numerical representations suitable for machine learning models, several complementary acoustic features were extracted. Since classical machine learning algorithms require fixed-length input vectors, statistical summaries (mean and variance) of frame-level features were computed for each audio file.

The final dataset contains **46 features per sample**.

3.1 Mel-Frequency Cepstral Coefficients (MFCCs)

Twenty Mel-Frequency Cepstral Coefficients (MFCCs) were extracted from each recording. For each coefficient, both the mean and variance were computed across time, resulting in **40 MFCC-based features**.

MFCCs are widely used in speech processing because they approximate human auditory perception and effectively capture the spectral envelope of speech. Since different languages exhibit distinct phonetic structures, vowel systems, and articulation patterns, MFCCs provide strong discriminative information for language identification tasks.

3.2 Spectral Centroid

The spectral centroid describes the “center of mass” of the frequency spectrum and reflects the perceived brightness of a sound. The mean and variance of this feature were included.

This feature helps capture differences in frequency distribution and articulation characteristics between languages.

3.3 Zero Crossing Rate (ZCR)

Zero Crossing Rate measures how frequently the signal waveform changes sign. The mean and variance were computed.

ZCR provides information about the presence of high-frequency components and fricative sounds, which may vary across languages due to differences in phonetic composition.

3.4 Root Mean Square (RMS) Energy

RMS energy represents the overall loudness and energy dynamics of speech. Its mean and variance were extracted.

This feature captures variations in speech rhythm, stress patterns, and prosodic characteristics, which can contribute to language discrimination.

3.5 Final Feature Representation

Each audio sample was ultimately represented by:

- 40 MFCC-based features
- 2 Spectral Centroid features
- 2 Zero Crossing Rate features
- 2 RMS Energy features

for a total of **46 features per recording**.

By combining spectral, temporal, and energy-based characteristics, the selected feature set provides a comprehensive representation of speech signals, making it suitable for both classification and clustering of multilingual audio data.

Classification

The goal of this stage was to train supervised machine learning models to automatically identify the spoken language of each audio recording using the extracted acoustic features.

4.1 Data Splitting

The dataset was divided into training and testing sets using an 80%–20% split. Stratified sampling was applied to ensure that all four languages (German, Italian, Korean, and Spanish) were equally represented in both subsets.

The final test set consisted of 144 samples (36 per language), ensuring a balanced evaluation.

4.2 Feature Scaling

Since many machine learning algorithms are sensitive to feature magnitude, standardization was applied using z-score normalization. Each feature was transformed to have zero mean and unit variance based on the training set statistics.

Feature scaling is particularly important for distance-based and margin-based classifiers such as K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Logistic Regression, and Multi-Layer Perceptron (MLP).

4.3 Classification Models

Five classification algorithms were implemented and compared:

- Logistic Regression
- Support Vector Machine (RBF Kernel)
- Multi-Layer Perceptron (MLP)
- K-Nearest Neighbors (K=5)
- Random Forest (50 trees)

Logistic Regression

Logistic Regression is a linear classifier that models class probabilities using a decision boundary in feature space. It serves as a strong baseline and performs well when classes are linearly separable.

Support Vector Machine (RBF Kernel)

SVM with a radial basis function (RBF) kernel was used to capture potential non-linear relationships between features. SVMs are known to perform well in high-dimensional feature spaces.

Multi-Layer Perceptron (MLP)

A neural network with one hidden layer (50 neurons) and ReLU activation was implemented. MLP can model complex non-linear patterns and interactions between acoustic features.

K-Nearest Neighbors (KNN)

KNN is a distance-based classifier that assigns labels based on the majority class among the 5 nearest neighbors. It is simple yet effective when similar samples cluster closely in feature space.

Random Forest

Random Forest is an ensemble method consisting of multiple decision trees. It captures non-linear feature interactions and is robust to noise and overfitting.

4.4 Evaluation Metrics

Model performance was evaluated using:

- Accuracy
- Precision (Weighted Average)
- Recall (Weighted Average)
- F1-Score (Weighted Average)
- Confusion Matrix

Weighted averages were used due to their robustness, although the dataset is balanced.

4.5 Results

The classification results on the test set are summarized below:

- Logistic Regression: Accuracy = 1.00
- SVM (RBF Kernel): Accuracy = 1.00
- MLP: Accuracy = 1.00
- KNN (5 Neighbors): Accuracy = 1.00
- Random Forest: Accuracy = 0.99

All models except Random Forest achieved perfect performance with precision, recall, and F1-score equal to 1.00 for all languages.

4.6 Confusion Matrix Analysis

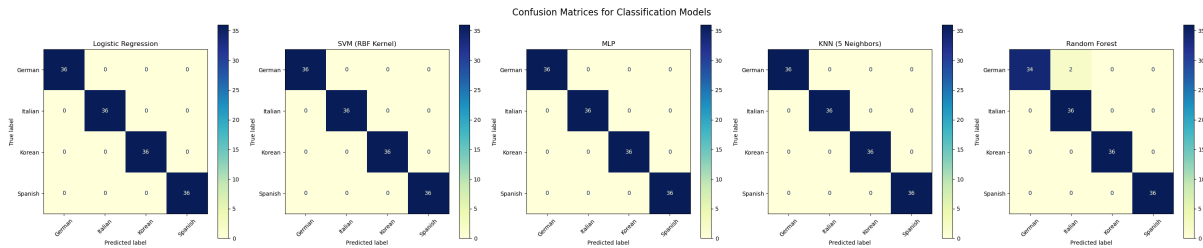


Figure 1: Confusion Matrix for different models

The confusion matrices show that Logistic Regression, SVM, MLP, and KNN correctly classified all 144 test samples without any misclassification.

Random Forest misclassified a small number of German samples as Italian, resulting in a slightly lower recall for German (0.94). However, overall performance remained extremely high.

Notably:

- Korean was perfectly classified across all models.
- Spanish also showed complete separability.
- Minor confusion occurred only between German and Italian in the Random Forest model.

This suggests that Romance and Germanic languages may share some overlapping acoustic properties, though the extracted features were generally sufficient to distinguish them.

4.7 Model Comparison and Discussion

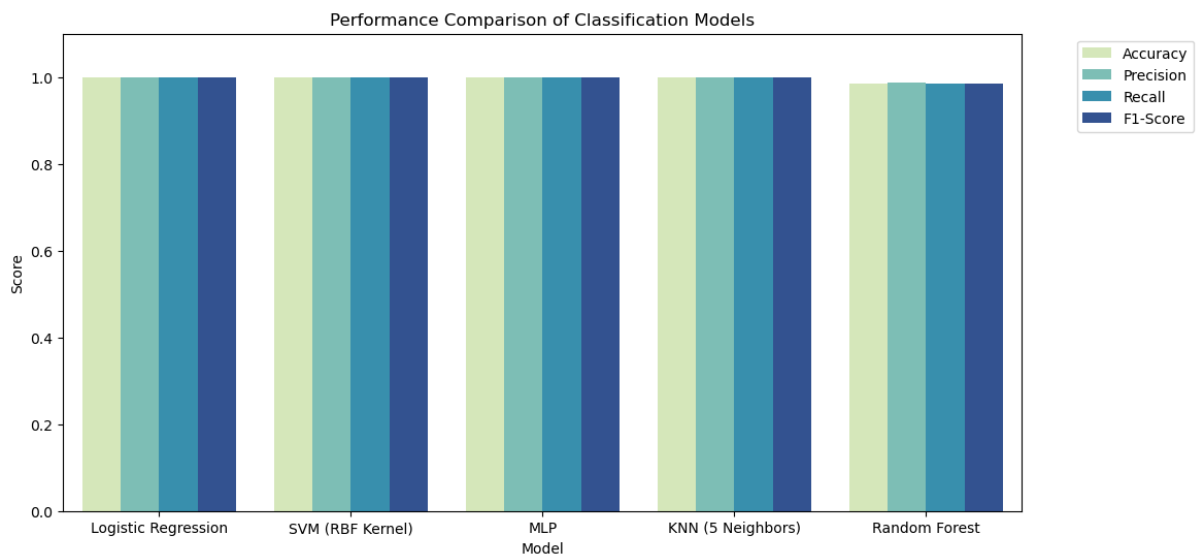


Figure 2: Performance Comparison of Classification Models

The near-perfect performance across multiple fundamentally different models (linear, kernel-based, neural network, distance-based, and ensemble methods) indicates that the extracted feature space provides strong discriminative power for language identification.

Several observations can be made:

- The perfect performance of Logistic Regression suggests that the classes may already be linearly separable in the 46-dimensional feature space.
- The similar performance of SVM and MLP confirms that non-linear modeling does not significantly improve results, further indicating strong feature separability.
- KNN achieving perfect accuracy suggests that samples from the same language cluster closely in feature space.
- Random Forest's slight misclassification may be due to tree-based partitioning variability rather than inherent feature overlap.

Overall, the classification results demonstrate that the selected acoustic features (MFCC, spectral centroid, ZCR, and RMS) are highly effective for multilingual speech identification in this dataset.

4.8 Interpretation of Results

The extremely high performance may be attributed to:

- Clear phonetic and prosodic differences between Korean and European languages.
- Distinct spectral characteristics captured by MFCC features.
- Balanced dataset and consistent preprocessing.
- Sufficient feature dimensionality to separate languages.

These findings confirm that traditional machine learning methods combined with well-engineered acoustic features can achieve excellent performance in spoken language classification tasks.

Clustering

In addition to supervised classification, an unsupervised analysis was conducted to investigate whether multilingual speech samples naturally group according to their language characteristics in the extracted acoustic feature space. Unlike classification, clustering algorithms do not use ground-truth labels during training. Instead, they attempt to discover inherent structural patterns in the data. This allows us to evaluate how well the extracted features capture language-specific properties without explicitly guiding the model.

In this project, four clustering algorithms were implemented and compared: K-Means, Hierarchical (Agglomerative) Clustering, Gaussian Mixture Models (GMM), and DBSCAN. The objective was to determine whether speech signals from German, Italian, Korean, and Spanish form distinct clusters in feature space and to evaluate which algorithm best captures this structure.

5.1 Data Preparation for Clustering

Before applying clustering algorithms, it was necessary to properly preprocess the data. Since clustering methods such as K-Means, Hierarchical Clustering, and DBSCAN rely heavily on distance computations, differences in feature scales can significantly distort the results. Therefore, all features were standardized using the StandardScaler transformation, which ensures zero mean and unit variance for each feature. This guarantees that no single feature dominates the distance metric due to scale differences.

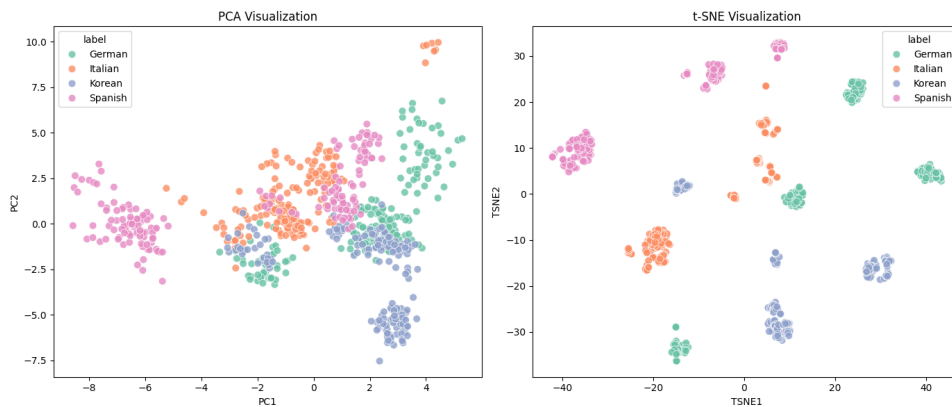


Figure 3: PCA and t-SNE Visualization

To visually analyze cluster structure, both PCA (2D projection) and t-SNE were used for visualization. PCA provides a linear projection preserving global variance structure, whereas t-SNE captures local neighborhood relationships and often reveals cluster separation more clearly.

5.2 K-Means Clustering

K-Means clustering partitions the data into k clusters by minimizing the within-cluster sum of squared distances (inertia). The algorithm assumes spherical clusters and relies on Euclidean distance to assign samples to cluster centroids.

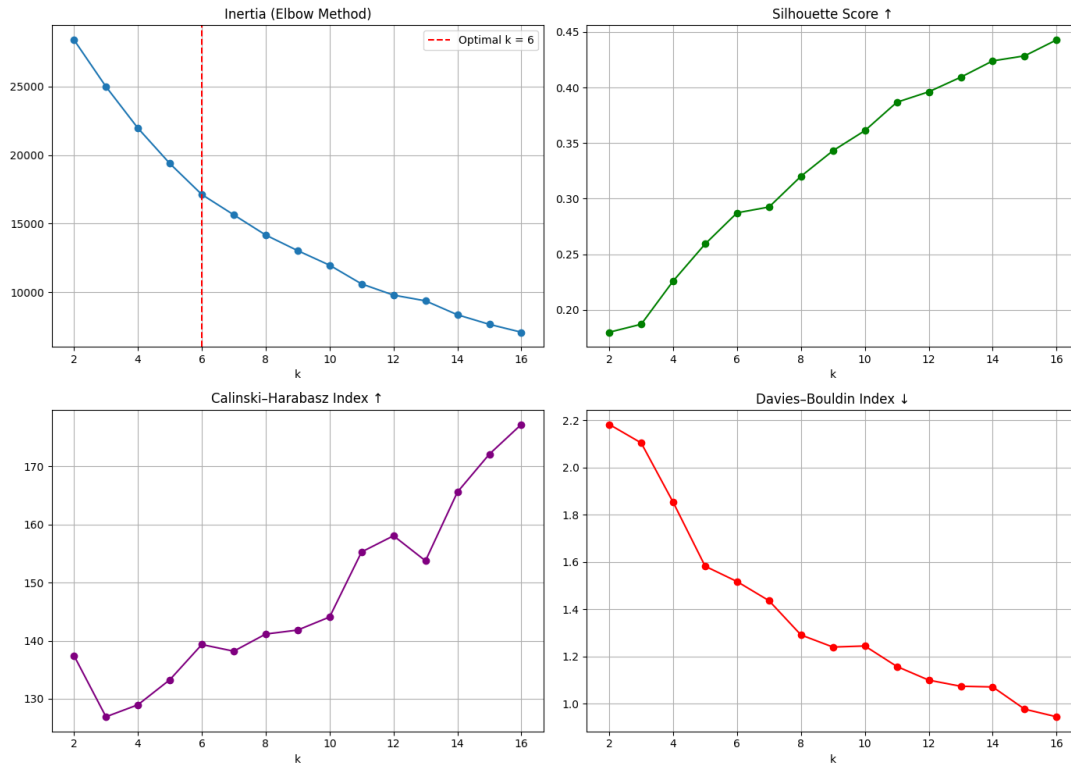


Figure 4: Evaluation Metrics

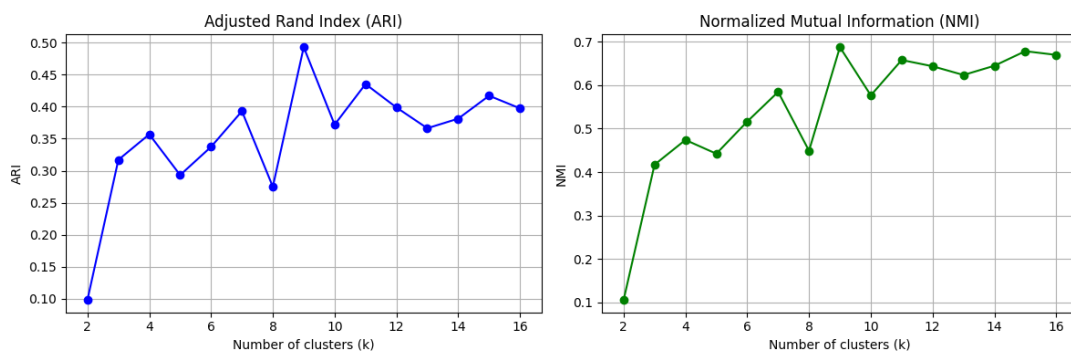


Figure 5: Evaluation Metrics to find Optimal K

To determine the optimal number of clusters, values of k from 2 to 16 were evaluated. Several internal validation metrics were analyzed, including inertia (Elbow Method), Silhouette Score, Calinski–Harabasz Index, and Davies–Bouldin Index. The Elbow Method, combined with automatic knee detection, suggested an optimal value of $k = 6$. Although silhouette scores

continued to increase gradually for larger values of k , selecting very large numbers of clusters risks over-segmentation and reduces interpretability.

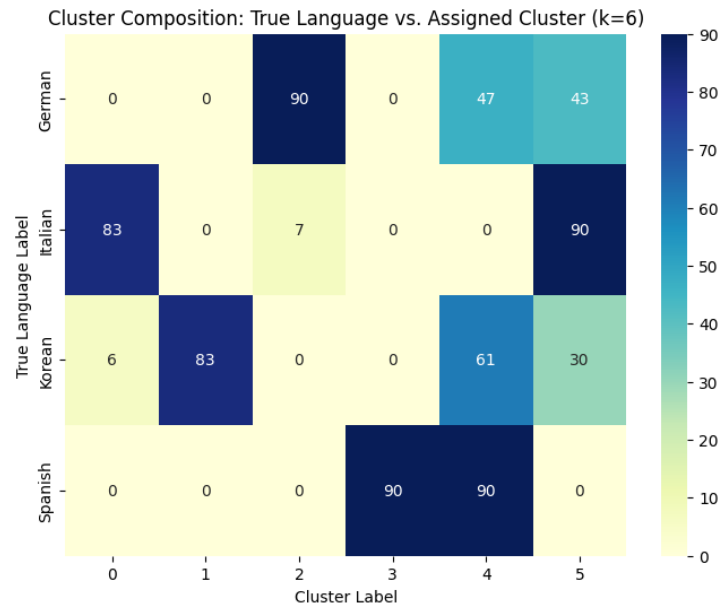


Figure 6: True Language vs. Assigned Cluster - K-Mean

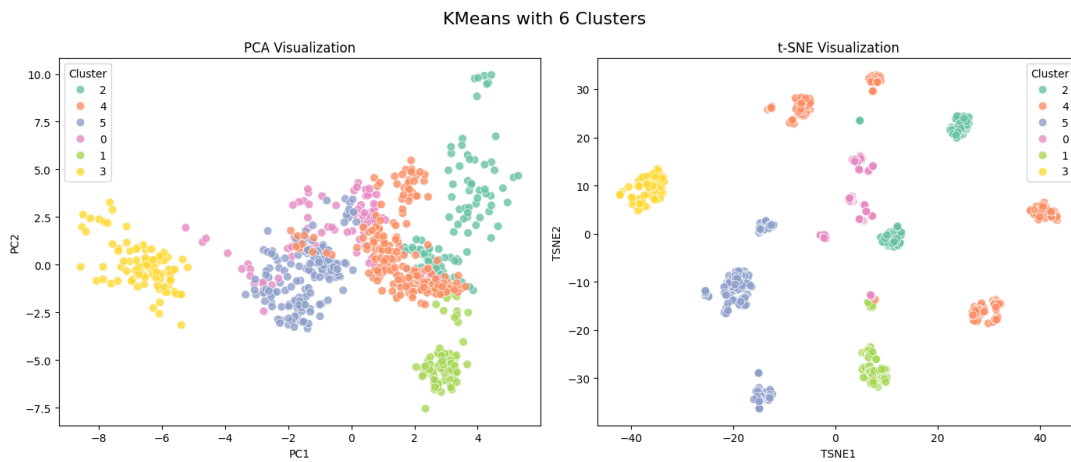


Figure 7: Visualization using PCA and t-SNE for K-Mean

For $k = 6$, K-Means achieved a Silhouette Score of 0.2873. External evaluation metrics comparing clusters with true language labels showed an Adjusted Rand Index (ARI) of approximately 0.34 and a Normalized Mutual Information (NMI) of approximately 0.52. The cluster purity was 0.7306, meaning that roughly 73% of samples were assigned to clusters dominated by a single language.

Visualization using t-SNE revealed relatively compact and separated clusters, although not perfectly aligned with the four true languages. In several cases, individual languages were

split into multiple clusters, suggesting that intra-language acoustic variability exists. Overall, K-Means provided strong baseline performance and demonstrated that meaningful structure exists in the feature space.

5.3 Hierarchical (Agglomerative) Clustering

Hierarchical clustering was implemented using Agglomerative Clustering with Ward linkage. Ward's method merges clusters in a way that minimizes the increase in within-cluster variance, often producing compact and well-separated clusters similar to K-Means.

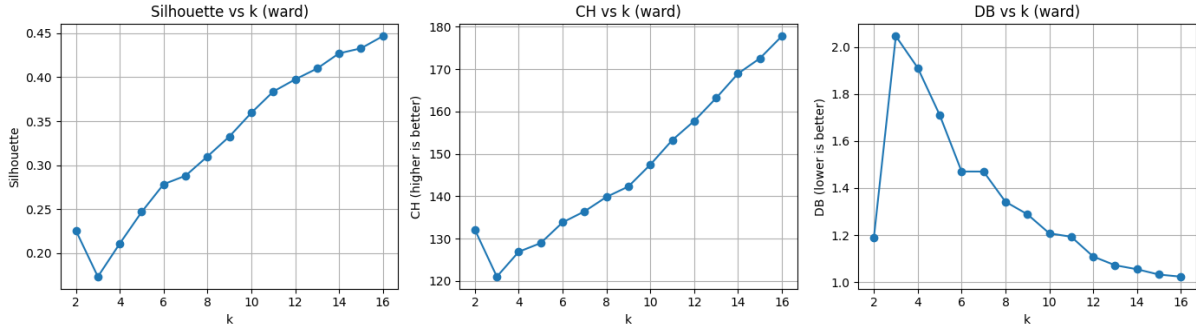


Figure 8: Evaluation Metrics to find Optimal K - Agglomerative Clustering

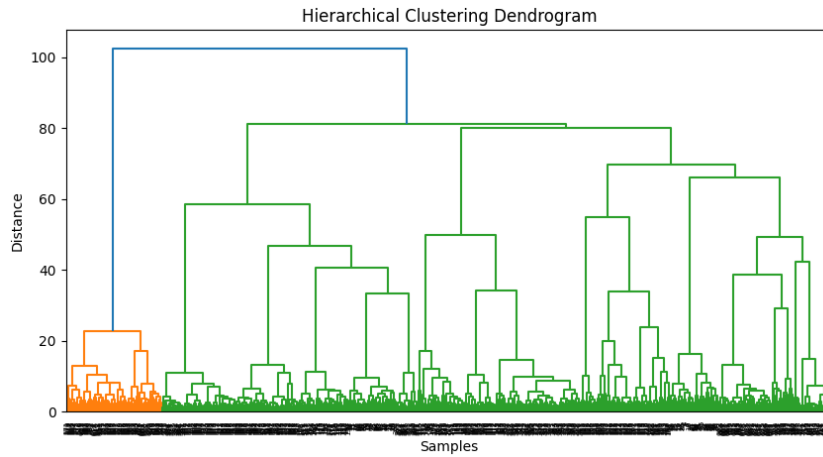


Figure 9: Hierarchical Clustering Dendrogram with $K = 2$

The dendrogram provided valuable insight into the hierarchical structure of the data. A large structural split was observed at $k = 2$, indicating a strong high-level division in the dataset. However, the silhouette score for $k = 2$ was relatively low (0.225), suggesting that although the macro-structure was clear, the clustering was too coarse to capture meaningful language distinctions.

A more balanced structure emerged around $k = 6$, where internal metrics showed improvement while maintaining interpretability. Therefore, $k = 6$ was selected for evaluation.

For this configuration, the silhouette score was 0.2782. The ARI was approximately 0.41 and the NMI approximately 0.58, both higher than those achieved by K-Means. The cluster

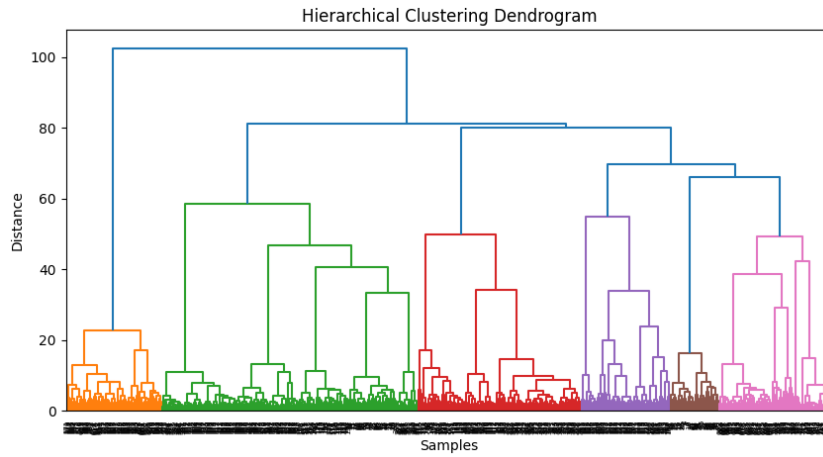


Figure 10: Hierarchical Clustering Dendrogram with $K = 6$

purity reached 0.7486, which was the highest among all tested methods. This indicates that hierarchical clustering achieved the best alignment between clusters and true language labels.

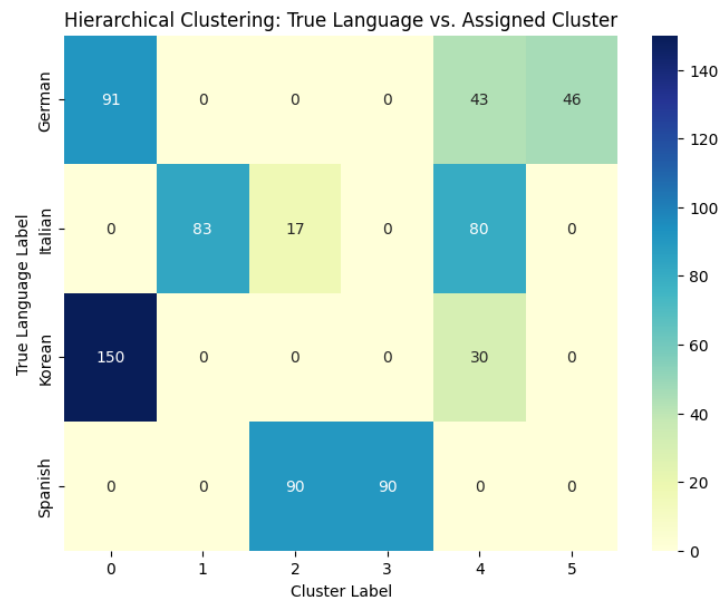


Figure 11: True Language vs. Assigned Clusters - Hierarchical

Visual inspection confirmed that clusters were slightly better separated compared to K-Means. These results suggest that Ward linkage effectively captures the variance structure of the PCA-reduced feature space.

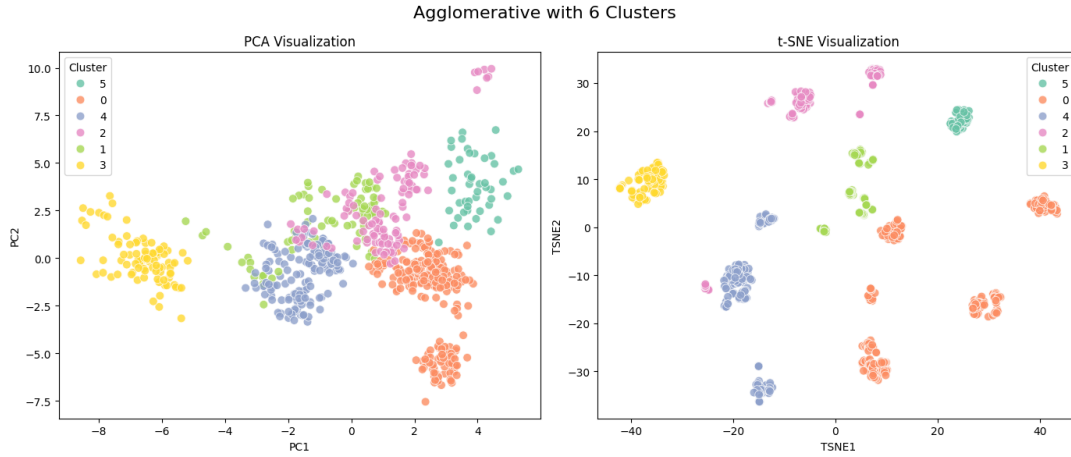


Figure 12: Visualization using PCA and t-SNE for Hierarchical Algorithm

5.4 Gaussian Mixture Model (GMM)

Gaussian Mixture Models provide a probabilistic approach to clustering by assuming that the data is generated from a mixture of Gaussian distributions. Unlike K-Means, which assigns samples to clusters deterministically, GMM allows soft probabilistic assignments and can model elliptical cluster shapes.

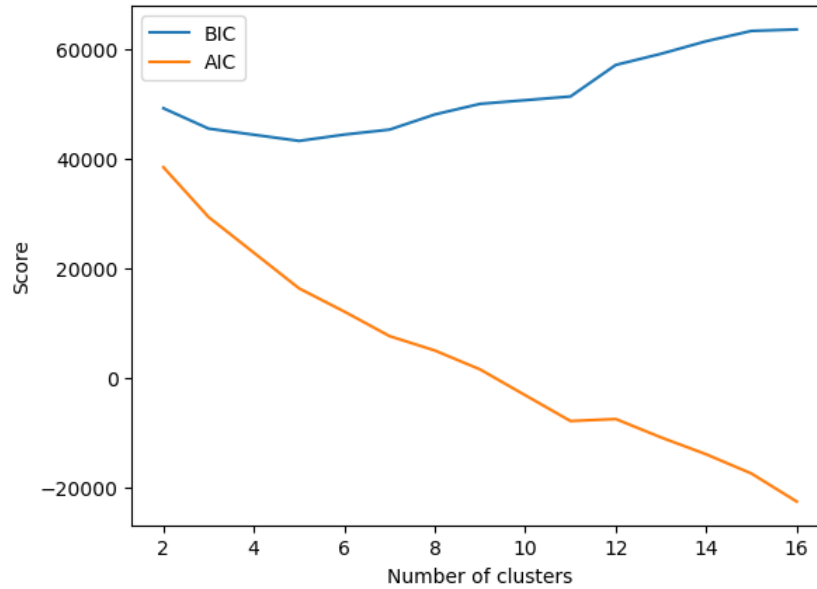


Figure 13: Evaluation Metrics for Different K - GMM

To determine the optimal number of components, the Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC) were evaluated for 2 to 16 components. Based on these curves, five components were selected.

With five components, the GMM achieved a silhouette score of 0.2470. The ARI was ap-

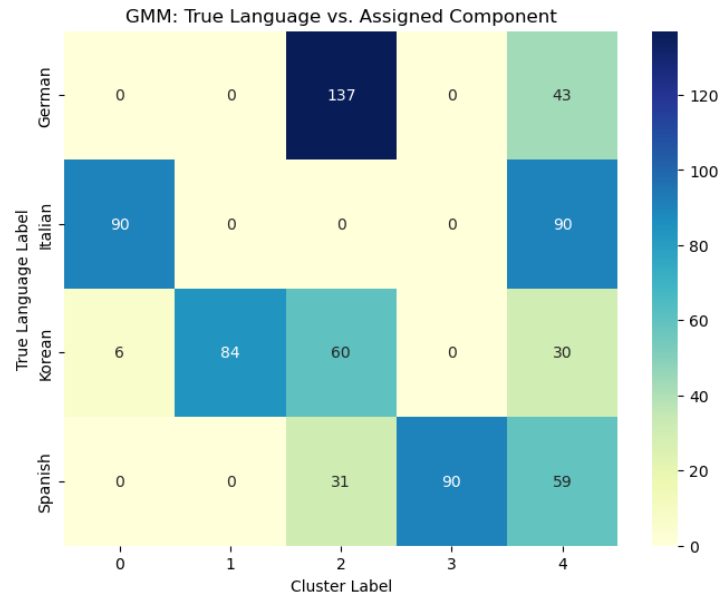


Figure 14: True Language vs. Assigned Clusters - GMM

proximately 0.30 and the NMI approximately 0.46. The cluster purity was 0.6819, lower than both K-Means and Hierarchical clustering.

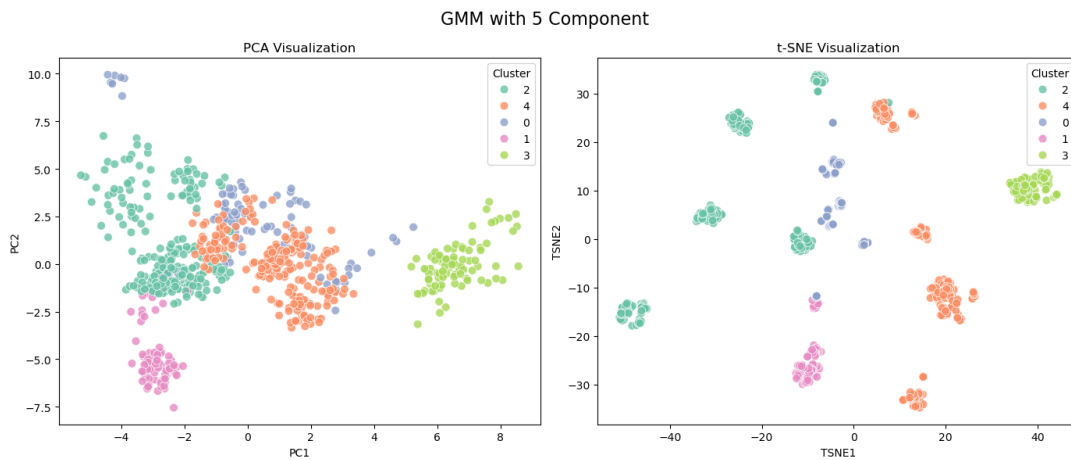


Figure 15: Visualization using PCA and t-SNE for GMM Algorithm

Although GMM provides greater flexibility in modeling cluster shapes, the speech feature distributions did not appear to benefit significantly from this probabilistic modeling. Overlapping Gaussian components may have reduced cluster clarity, leading to lower alignment with the true language labels.

5.5 DBSCAN

DBSCAN is a density-based clustering algorithm that does not require pre-specifying the number of clusters. It groups samples based on density connectivity and can identify noise points. This makes it attractive for datasets with irregular cluster shapes.

Hyperparameters were tuned using k-distance analysis and knee detection. The selected parameters were $\epsilon = 6.677$ and $\text{min_samples} = 10$. With these values, DBSCAN identified three clusters and labeled approximately 1.39% of the data as noise.

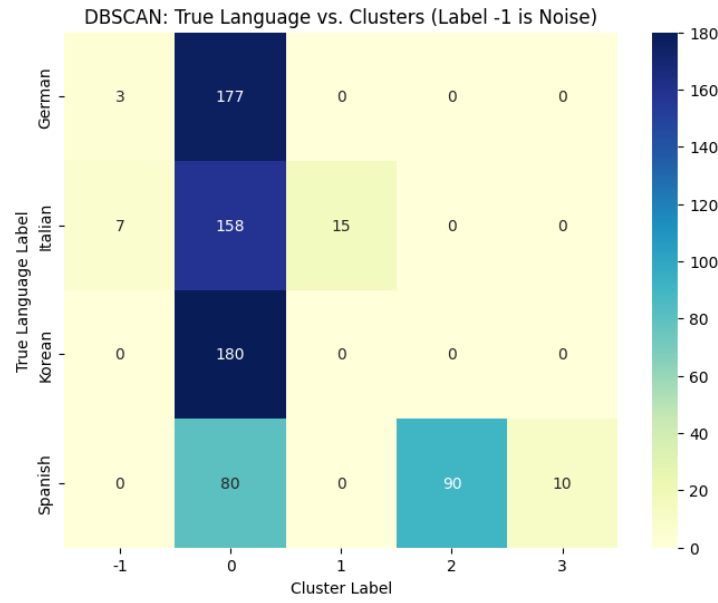


Figure 16: True Language vs. Assigned Clusters - DBSCAN

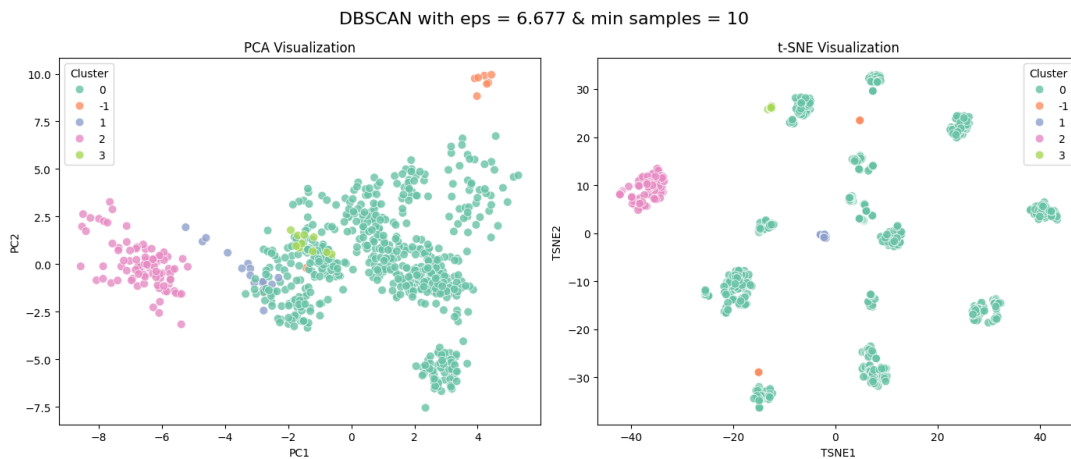


Figure 17: DBSCAN with $\text{eps} = 6.677$ and $\text{min sample} = 10$

After excluding noise points, the silhouette score was 0.4194. However, external evaluation metrics were considerably lower than other methods, with ARI around 0.08, NMI around 0.27,

and cluster purity only 0.4194.

These results indicate that the density distribution in the feature space does not strongly support density-based separation of languages. DBSCAN grouped data into broad density regions rather than language-specific clusters.

5.6 Comparative Analysis

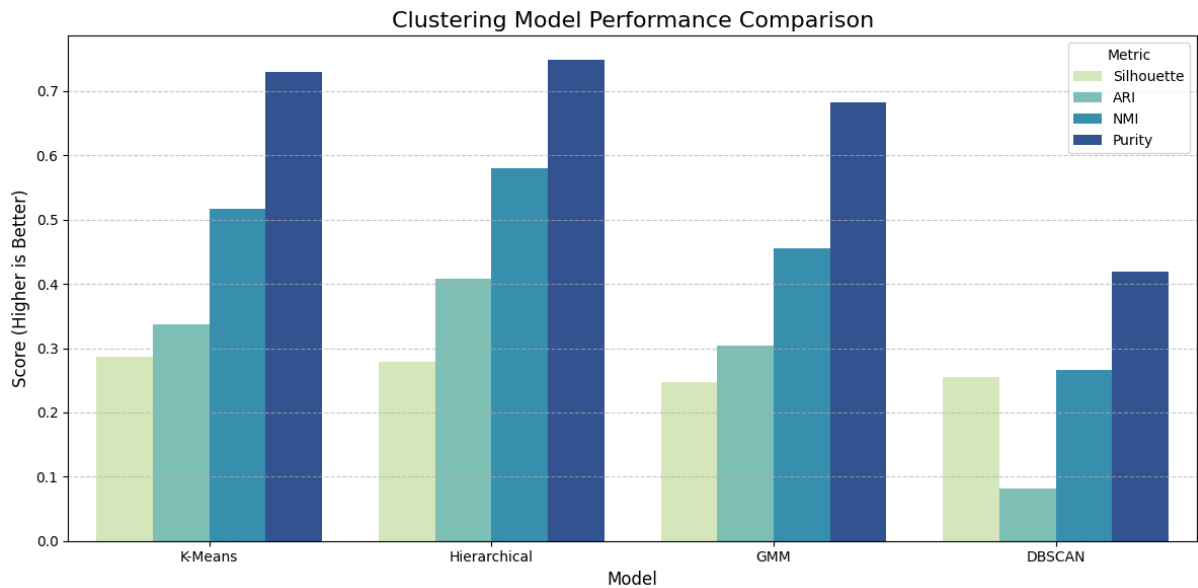


Figure 18: Clustering Model Performance Comparison

Comparing the four clustering methods reveals clear differences in performance. Hierarchical clustering achieved the highest purity and strongest alignment with true labels. K-Means performed similarly but slightly worse in external metrics. GMM showed moderate performance but did not outperform simpler distance-based approaches. DBSCAN was the weakest method for this dataset.

Interestingly, several methods suggested optimal cluster numbers greater than four, even though the dataset contains four languages. This indicates the presence of sub-structure within individual languages, likely caused by speaker variability, pronunciation differences, or acoustic diversity.

5.7 Overall Conclusion for Clustering

The clustering analysis demonstrates that meaningful structure exists in the multilingual speech feature space. However, languages are not perfectly separable using unsupervised learning

alone. Overlapping acoustic characteristics between certain languages prevent complete separation.

Among the tested algorithms, Hierarchical Clustering with Ward linkage provided the best overall performance, achieving the highest cluster purity and strongest alignment with true language labels. K-Means was a competitive and computationally efficient alternative. GMM offered probabilistic modeling but did not improve results. DBSCAN was not well suited for this dataset due to insufficient density separation.

These findings highlight an important insight: while supervised classification can achieve high accuracy by learning explicit decision boundaries, unsupervised clustering reveals that language distinctions in acoustic feature space are partially overlapping rather than perfectly distinct.